

COLLEGE MANAGEMENT ERP SYSTEM

Project Documentation

1. Introduction

The College Management ERP System is a web-based application designed to manage academic and administrative activities of a college. It provides role-based access to users such as Super Admin, Admin (HOD), Faculty, and others.

The system helps in managing students, departments, academic structure, attendance, and marks efficiently.

2. Objectives

- To automate college management processes
 - To maintain centralized data
 - To implement role-based access
 - To track students using batch, year, and semester
 - To generate reports and analytics
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3. Technologies Used

Frontend:

- React.js (Vite)
- Tailwind CSS

Backend:

- Node.js
- Express.js

Database:

- MySQL

Other Tools:

- JWT Authentication
 - Bcrypt
 - GitHub
 - Ngrok
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4. System Architecture

The system follows a 3-tier architecture:

- Frontend (React UI)
 - Backend (Node.js APIs)
 - Database (MySQL)
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5. Roles in System

- Super Admin (Principal)
 - Admin (HOD)
 - Faculty
 - Supporting Staff
 - Accountant
 - Administrator
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6. Academic Structure

The system follows this hierarchy:

Course → Department → Batch → Year → Semester → Section

Example:

- Batch: 2021–2025
 - Year 1 → Semester 1, 2
 - Year 2 → Semester 3, 4
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7. Super Admin Features

- Login with phone and password
- Create courses (B.Tech, MBA)
- Create departments (CSE, ECE)
- Create academic batches (2021–2025)
- Manage sections based on year & semester
- Create HODs
- Manage subjects with credits
- Define fee structure

- Create academic calendar
 - Post announcements
 - View reports and analytics
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8. Key Features

- Role-based authentication
 - Academic batch tracking
 - Dynamic year & semester system
 - Centralized data management
 - Scalable architecture
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9. Database Tables

- users
 - courses
 - departments
 - batches
 - sections
 - students
 - faculty
 - subjects
 - attendance
 - marks
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10. Advantages

- Reduces manual work
 - Improves accuracy
 - Easy tracking of students
 - Centralized system
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11. Future Enhancements

- Student login

- Mobile app
 - Online fee payment
 - Notifications (SMS/Email)
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12. Conclusion

The system provides an efficient way to manage college operations and follows real-world ERP architecture.